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PALEOMAGNETISM AND THE ALPINE TECTONICS OF EURASIA V

THE PALEOMAGNETISM OF SOME MESOZOIC AND CENOZOIC VOLCANIC ROCKS FROM THE LEBANON

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ABSTRACT

Gregor, C.B., Mertzman, S., Nairn, A.E.M. and Negendank, J., 1974. Paleomagnetism and the Alpine tectonics of Eurasia V. The paleomagnetism of some Mesozoic and Cenozoic volcanic rocks from the Lebanon. Tectonophysics, 21: 375—395.

A sampling of Mesozoic and Tertiary basalts in Lebanon yielded the following information:

Age	D	I	α_{95}	Pole position		d_p	d_m
Upper Jurassic	95	+21	10.6	114E	2N	5.9	11.2
				66W	2S		
Lower Cretaceous	122	+ 2	9.0	105E	25S	4.5	9.0
				75W	25N		
Upper Pliocene	2	+46	7.7	169E	88N	6.3	9.8
				11W	88S		

These results confirm and amplify earlier work by Van Dongen et al., and can be interpreted as indicating a net anticlockwise rotation of Lebanon relative to the African tectonic plate amounting to about 70° during the Late Jurassic—Pliocene interval. This could have resulted from differential movement between the African and European plates as they made way for the growing Atlantic Ocean.

INTRODUCTION

Throughout most of Phanerozoic history an important seaway has linked the Mediterranean region with the tropical provinces of the Far East. Clear evidence of its existence is found in the Mesozoic rocks of the Middle East; and one of the most important of geological events, whatever its cause, was

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the interruption of this seaway sometime in the Late Mesozoic or the Cenozoic. The isolation of the Mediterranean Sea from the Tethys was but one consequence of extreme tectonic activity in the whole Tethyan region in Late Mesozoic and Cenozoic times. The latest phase in this history has been dealt with in some detail by McKenzie (1972).

Traditionally, tectonic activity has been considered separately from the large-scale movements associated with continental drift; for until recently attention on the latter was centered around the regions of separation rather than on the regions where continental blocks have approached one another. The two are now unified by the sea-floor spreading–plate tectonics concept. Viewed from the plate tectonics standpoint the Mediterranean is of paramount importance for the possibly unique opportunity it affords of following in detail the mutual approach of two opposed continental blocks.

Paleomagnetic data from numerous studies in the western Mediterranean have commonly been interpreted in terms of rotational or translational movements with respect to northern Europe (e.g. Van der Voo, 1967; Nairn and Westphal, 1968; Zijderveld et al., 1970). The tectonic traits of the region have been discussed by Argand (1924), by Hsü (1971), and in numerous papers by Glangeaud and his co-workers (Glangeaud et al., 1967); and Smith (1971) has reconciled paleomagnetic poles for some of the microcontinents with a least-squares fit of their 1000-m isobath. Rather less information is available from the eastern Mediterranean, mainly because of poorer geological control. There is nonetheless impressive evidence of translational motion along the Pamphlagonian (trans-Anatolian) Fault (Pavoni, 1964), and there seem to be mantle fragments in the crust of both Turkey and Cyprus (Vine and Moores, 1969). Recently McKenzie (1970) and McKenzie et al. (1970) have discussed the application of plate tectonics to the Middle East, and their conclusions seem to be supported by seismic studies (Canitez and Toksöz, 1970). Lort (1971) summarizes the available information and develops the idea of a number of small plates between the African and European plates.

The Near East countries bordering the Mediterranean are generally regarded as part of the Afro-Nubian shield. If they are, the late history of this part of the Tethys must be intimately connected with the behavior of the African plate; moreover, if rotation of the European microcontinents (Spain, Sardo-Corsica, Italy and Greece) has resulted, as Hsü (1971) suggests, from movement of Africa relative to Europe, there is every reason to look for similar deformation on the northern fringe of the African plate. While paleomagnetic results are not particularly sensitive indicators of "translational" motion*, they are extremely sensitive to rotation. Up to the present there

*No translations are possible on a sphere. The "translations" commonly mentioned in a geotectonical context are really rotations about axes emerging far from the crustal areas whose movements are referred to them.

have been relatively few studies on rocks from Middle Eastern countries (see for instance Van Dongen et al., 1967; Gregor and Zijderveld, 1964; Van der Voo and Van der Kleijn, 1970). This paper reports the results of a collection made in 1968, and amplifies the results of Van Dongen et al. (1967).

GENERAL GEOLOGY AND PETROGRAPHY

The Lebanon (Fig. 1) is a country with a varied relief reflecting its general geologic structure. The most conspicuous features are two ranges, the Lebanon and the Antilebanon, which trend approximately north-south and flank the northern extension of the Dead Sea rift. The erosion of large-amplitude open folds, which are interrupted by the great rift boundary fault, has exposed more than 5,000 m of Mesozoic and younger rocks, predominantly carbonates. The lower part of the succession is best exposed in central and northern Lebanon; the younger beds are best seen in the southern part of the country.

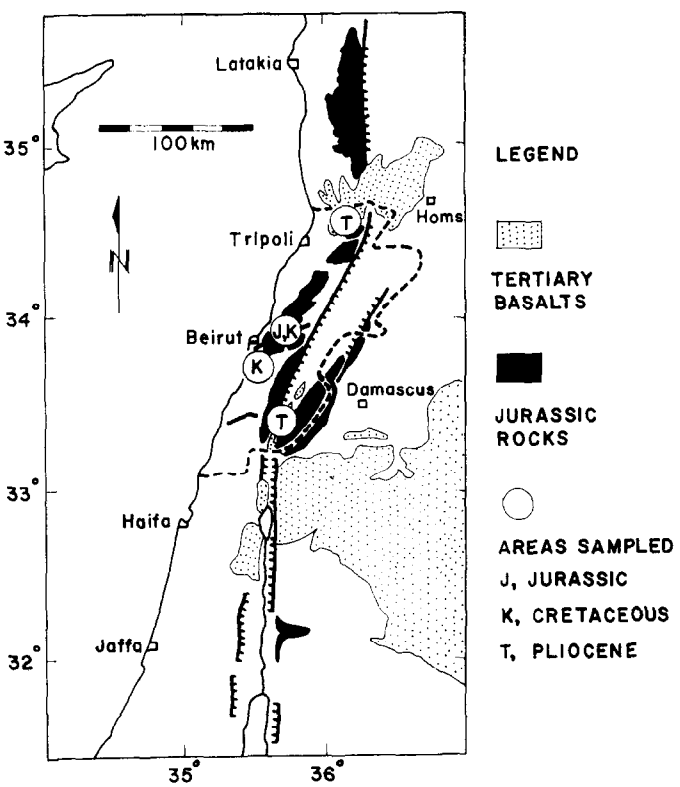


Fig. 1. Sketch map of the Lebanon and adjacent regions showing locations of the areas sampled.

Early geological investigations were carried out by Fraas (1878), Diener (1886), Blanckenhorn (1914) and Zumoffen (1926); these were followed by Dubertret (1955) and Wolfart (1967). The Mesozoic section, made up predominantly of limestones, contains several volcanic horizons near the Jurassic—Cretaceous boundary (Fig. 2). A further significant phase of volcanism occurred during Neogene (Pliocene) times. The thick limestones with their benthonic fauna indicate that during much of Mesozoic time the region was a slowly subsiding marginal basin. The subsidence was interspersed with periods of uplift and even local emergence; the most important of these, at the end of the Jurassic, is recorded in a Lower Cretaceous sandstone (Grès du Liban) which rests (in most places unconformably) upon Jurassic rocks. The first stirrings of orogeny are found in the Turonian limestones, whose variable thickness indicates accentuated submarine relief. These movements became progressively more insistent and culminated with the development of Lebanon's main structural features in Neogene times. Rising mountains supplied gravel which poured into the rift to the east, while in the downwarped coastal regions carbonates and clays accumulated. The later stages of this tectonic activity saw outpourings of basalt in the northern and southern parts of the country. Across the southern boundary of the Lebanon volcanic activity has continued down to historic times.

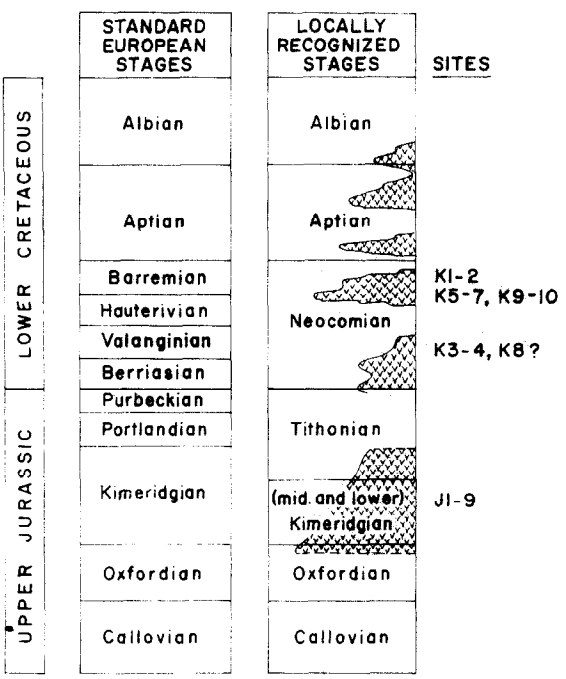


Fig. 2. Approximate stratigraphic positions of the Jurassic and Cretaceous basalts in Lebanon, and horizons sampled. (Sites numbered as in Tables I and II.)

The ages of the basalts, of prime concern to this paper, can be defined within fairly close stratigraphic limits (see Fig. 2). The oldest horizon, which consists of some 200 m of basalt flows, pyroclastics and laterites, overlies a limestone with an Oxfordian fauna and in turn is overlain by limestones regarded as Tithonian because they contain fossils of *Berriasella richteri* Oppel (Arkell, 1956, p. 296). Dubertret (1955) correlates the base of the volcanic zone with a limestone bearing an Upper Oxfordian fauna in the Antilebanon. This volcanic episode therefore occurred sometime between the Middle Oxfordian and the Late Tithonian; it is commonly attributed to the Kimeridgian.

More basalt erupted during deposition of the Grès du Liban. This sandstone, barren of guide fossils, is overlain by limestones with a Lower Aptian fauna. It is therefore almost certainly Neocomian but cannot be dated more precisely. Further eruptions occurred during the Aptian on into Early Albian time but they were more local in character, and volcanic horizons seldom more than a few metres thick are all that is known of them. The products of more prolonged volcanism during Neogene times are found in northern Lebanon. According to Guerre (1968) three distinct formations can be recognized:

(1) β_p 3-subaerial basalt flows of Plaisancian age at the base, probably extending into the Pleistocene;

(2) β_p 2-pyroclastics and volcanic breccias of Plaisancian age;

(3) β_p 1-submarine basalts interfingering with Plaisancian marls.

The thickest and best exposed of these horizons is β_p 3. The stratigraphic relationships of the Neogene basalts in the southern part of the country are less clear. These lavas, which rest uncomformably on Neogene conglomerates in the rift, have been broken by prolonged erosion into a chain of small, isolated outliers whose arrangement shows that they were deposited upon an earlier land surface long since destroyed. They are therefore regarded as Pliocene and the equivalents of β_p 3 in the north; however, the possibility of a Pleistocene age cannot be excluded.

In outcrop the basalts are generally greyish-black weathering to various shades of brown (or red when laterized). Most are compact (the Pliocene flows have vesicular tops) and porphyritic, with insets up to 5 mm in diameter. They have been studied by Kuttayneh (1967). They are composed of magnesian olivine and grey to purple titanaugite in varying proportions, calcic plagioclase, and minor Fe—Ti oxides, apatite, interstitial alkali feldspars and brown glass. The state of the olivine grains generally indicates the degree of alteration of the rock: in the Pliocene samples they are normally fresh or only marginally altered to iddingsite; in many of the Jurassic and Cretaceous rocks, on the other hand, they are completely altered except for the occasional core of a large phenocryst.

The opaque minerals occur as phenocrysts up to 1.3 mm in diameter; as microphenocrysts, and as groundmass constituents, making up in all from 1 to 8% by volume. Titanomagnetite and ilmenite predominate, and there are

TABLE I

Summary of ore mineralogy

Sites	Oxidation No.*		Mean grain diam. (μ)	Maghemite in titano-magnetite	Remarks
	M	*I*			
A. Jurassic					
1	1.10	1.0	needles	—	Ilmenite > tima
2	1.0	1.0	needles 130–735 by 31	—	Pyrite and chalcopyrite present
3	1.0	1.0	needles 458–571 by 11	—	Ilmenite >> tima, "intergrowths" of ilmenite—titanomagnetite
4	1.0	1.0	169	—	Ilmenite >> tima, "intergrowths" of ilmenite—titanomagnetite
5	—	—	—	—	Ore pigment too fine for identification
6	1.0	(1)	100	—	Pyrite also present
8	—	—	—	—	Ore pigment too fine for identification
9	1.05	1.0	242	x	Pyrite and chalcopyrite occur
B. Cretaceous					
1	1.6	1.0	233	x	Pyrite present
2	1.2	1.0	296	x	Pyrite and chalcopyrite, random lepidocrocite, tima/ilmenite = 1.1
3	1.8	1.0	276	x	Pyrite and chalcopyrite present; ilmenite—titanomagnetite intergrowths
4	6.0	3.0	needles	—	—
5	—	—	—	—	Ore pigment not identifiable
6	1.0	1.0	164	xx	Some granulation, pyrite and chalcopyrite, titanomagnetite completely replaced by maghemite
7	1.5	1.0	needles	xx	Some granulation, pyrite and chalcopyrite
8 sample 1	3.9	1.3	339	(x)	ilmenite >> tima
sample 2	1.0	1.0	183	x	pyrite present
9	1.1	1.0	needles 676 x 23	—	pyrite present

TABLE I (continued)

Sites		Oxidation No.*		Mean grain diam. (μ)	Maghemite in titano-magnetite	Remarks
		M	*I*			
10		1.3	1.0	needles 676 x 45	x	—
C. Pliocene						
1		2.41	(1)	155	(x)	pyrite, chromite present
2		1.0	(1)	87	x	99% titanomagnetite
3		4.5	2.07	130	—	—
4		4.3	2.03	167	—	ilmenite—titanomagnetite intergrowth, ilmenite/tima = 1 : 1
5		3.3	1.6	187	x	—
6		1.0	1.0	211	xx	pyrite present
7	sample 1	1.0	(1)	68	(x)	99% tima
	sample 2	1.1	2.0	68	—	—
	sample 3	4.59	2.03	needles	—	red around highly oxidized grains
8	sample 1	2.1	1.4	212	xx	—
	sample 2	3.3	2.25	231	—	—
9	sample 1	1.0	(1)	133	x	pyrite present
	sample 2	4.66	2.18	134	—	red around highly oxidized grains
10	sample 1	4.36	1.0	53	—	tima ilmenite
	sample 2	3.2	(1)	53	—	fluidal texture
11	sample 1	5.1	2.1	159	—	—
	sample 2	2.6	1.0	32	—	—
	sample 3	2.55	(1)	116	—	—
12		1.0	1.0	123	x	vermicular maghemite
13	sample 1	1.03	—	112	(x)	random granulation of tima
	sample 2	3.32	—	56	xx	—
14		3.46	1.7	109	—	—
15		1.0	(1)	74	xx	localized vermicular maghemite formation in reddened zones
16		4.9	(2)	131	—	tima ilmenite

* On the titanomagnetite scale (*M*) the oxidation numbers run from 1 to 6; on the ilmenite scale (*I*) they run from 1 to 3. The word "intergrowth" where it occurs in this table is intended as a descriptive term only.

some cubic and octahedral chromite crystals in the olivine insets. The oxidation state of the Fe—Ti oxides tends to parallel the state of preservation of the olivine. Where the latter is relatively fresh the opaques belong to oxidation classes 1—3 (Ade-Hall et al., 1968; Wilson et al., 1968); where it has been completely replaced, they belong to classes 4—6. The titanomagnetite grains have suffered some oxidation to maghemite, but this is patchy and irregular. Granulation, though present, is uncommon; and this, together with the small amount of maghemite, suggests that significant low-temperature alteration of the ore minerals has not occurred. No consistent correlation was observed between the degree of oxidation and the magnetic sign in those rocks (Cretaceous and Pliocene) where both normal and reversed directions were found. The opaque mineralogy is summarized in Table I.

PALEOMAGNETIC OBSERVATIONS

Oriented block samples of the Kimeridgian, Neocomian and Pliocene basalts were collected during 1968 from 35 sites (9 Jurassic, 10 Cretaceous and 16 Pliocene) whose distribution is shown in Fig. 3. An attempt was made to obtain at least six samples from each flow, but because of poor outcrop in some areas this was not always possible. From each block a single core was drilled which provided one or two cylinders. Initially two cores per site were magnetically cleaned by progressive demagnetization in alternating fields. The maximum field applied, 1200 Oe, was reached in five steps, and the magnetic vector was plotted after each step on a demagnetization diagram (Fig. 4 shows two examples; see As and Zijderveld, 1958). The field intensity beyond which the vector decayed towards zero intensity without further change in direction was judged to have removed all secondary components of magnetization; the remaining cores from a given site were then measured after demagnetization in that field and again at the next higher step. All the remanence measurements were made with a fluxgate spinner magnetometer (Foster, 1966), and the specimens were rotated about three axes while being demagnetized in a coil housed in a mu-metal non-magnetic room. The mean directions of magnetization (Fisher, 1953) were calculated on a Wang computer, and the direction regarded as best representing the original one at the time the rock was formed was selected by maximizing Fisher's parameter k (see Irving et al., 1961). These directions are listed in Table II and plotted in Fig. 6.

The intensity of natural remanent magnetization was found to be highly variable, ranging from $3 \cdot 10^{-5}$ to $300 \cdot 10^{-5}$ emu.cm^{-3}; usually it was consistent at any given site. There was no obvious correlation between alteration of the rock and its NRM intensity. Some of the demagnetization diagrams showed anomalies similar to those recorded by Zijderveld (1967): while in many of the samples the magnetic vector shrank towards the origin after removal of a secondary component (Fig. 4A), this did not always happen (Fig. 4B). The anomalous behaviour was generally seen when the intensity of

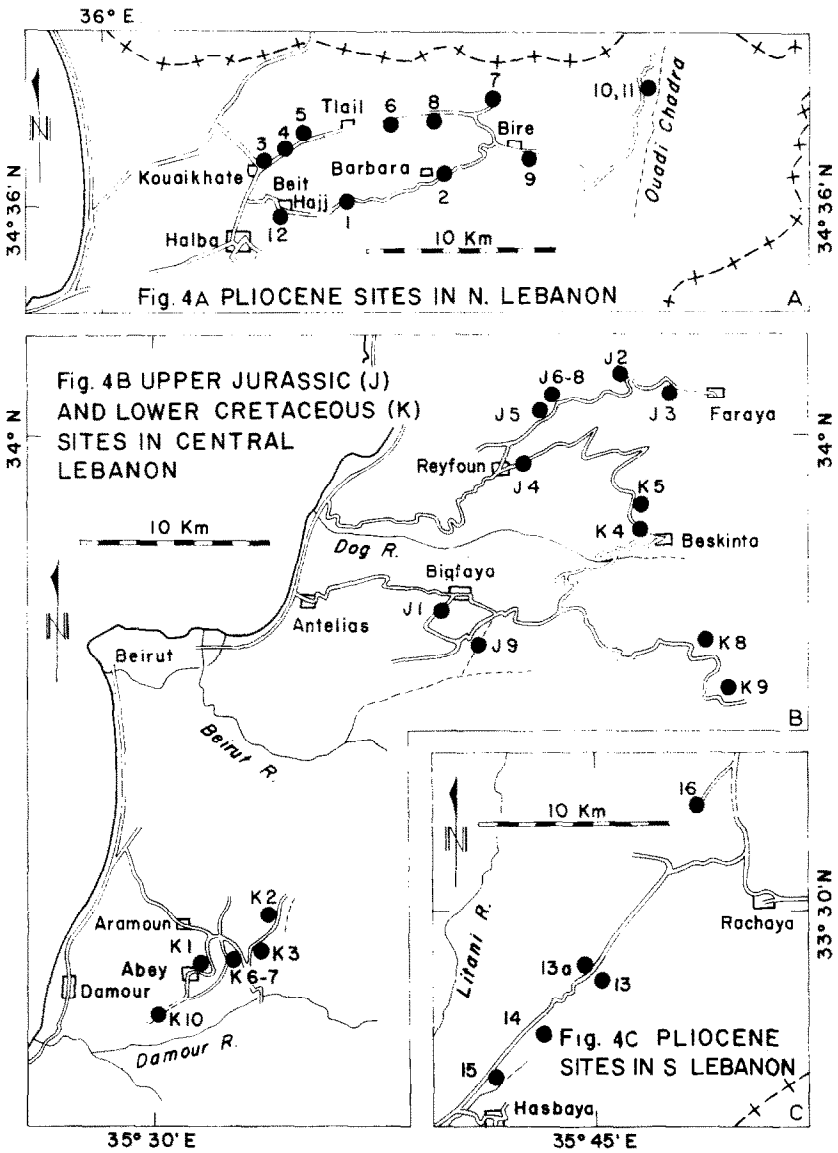


Fig. 3. Locations of the sampling sites. (Numbers as in Tables I and II.)

magnetization was low; the normalized demagnetization curves (Fig. 5) are similar for both normal and anomalous samples. As both kinds of behaviour were observed among samples from the same site, it is possible that the anomalous behaviour reflects measurement error in very weakly magnetized samples; alternatively there may be a secondary component of magnetization whose coercivity is close to that of the primary component. From a practical

TABLE II

Summary of paleomagnetic results from Lebanon

Sites		D	I	Sampling*	α_{95}	K	R	Demagn. (Oe)
A. Jurassic sites								
1	Bhersaf—Bhannes (132—138a)	80.0	+ 22.0	7/8	13.5	21	6.715	150
2	Mairouba (139—144)	98.4	+ 15.3	7/7	14.0	24	5.791	150
3	Hrajel (145—150)	95.0	− 2.5	6/6	8.8	59	5.915	150
4	Reyfoun (222—228)	95.8	+ 11.8	5/7	22.0	13	4.693	150
5	Boqa'ata (229—235)	62.8	+ 43.6	7/7	17.2	13	6.547	150
6	Boqa'ata (Tabrie Road) (237—242)	100.6	+ 38.7	6/6	8.1	69	5.928	150
7	Boqa'ata (Tabrie Road) dyke (243—245)	86.8	+ 36.9	4/4	19.5	23	3.871	150
8	Boqa'ata (Tabrie Road) dyke (248—254)	103.1	+ 27.4	6/7	35.5	5	4.892	300
9	Deir Mar Moussa (279—290)	99.8	+ 17.5	11/12	5.1	81	10.877	300
	Overall mean direction (Site 5 excluded)	95.0	+ 21.1	—	10.6	28	7.751	—
	Virtual pole position	114 E 66 W	2 N 2 S		dp = 5.9 dm = 11.2			
B. Cretaceous sites								
1	Abey (120—124a)	130.9	+ 4.6	5/5	18.8	18	4.773	150
2	Serjbel (125—131)	122.7	− 1.8	7/7	11.5	29	6.791	150
3	Nahr Chekfita (181—184) lava horizon well down in Neocomian sandstone	125.8	+ 2.0	4/4	5.2	314	3.990	300
4	Baskinta (212—217) lava well down in Neocomian	292.6	− 11.8	5/6	29.0	8	4.490	150
5	Baskinta (218—221) same horizon as 4, 1¼ km N	—	—	—	rock unstable	—	—	—
6	N Chouf near Dfoun (270—275)	131.8	+ 1.2	6/6	10.0	46	5.892	150

TABLE II (continued)

Sites		D	I	Sampling*	α_{95}	K	R	Demagn. (Oe)
7	Dfoun (291—297) same as 6, 500 m S	124.0	+ 6.5	5/7	15.0	27	4.852	300
8	Dara (298—306) Upper Juras. or Lower Cret. basalt	277.4	+ 6.4	8/9	6.7	69	7.898	150
9	Col of Zahle (307—315) Neocom. or L. Aptian near rift boundary fault	63.1**	− 3.1	7/9	20.1	10	6.396	300
10	N. Chouf, Kafr Matta (322—328)	132.1	+ 0.2	7/7	9.6	40	6.851	150
	Overall mean direction (Site 9 excluded)	122.3	+ 2.3	—	9.0	39	7.820	—
	Virtual pole position	105 E 75 W	25 S 25 N		dp = 4.5 dm = 9.0			
C. Pliocene sites								
1	Hidd, N Lebanon $\beta_p 3$ (111—114)	340.2	+ 32.9	4/4	4.8	373	3.992	150
2	Barbara, N Lebanon $\beta_p 3$ (115—119)	354.8	+ 52.7	4/5	8.9	109	3.972	300
3	Kouaikhate, N Lebanon $\beta_p 3$ (151—156)	8.4	+ 66.0	6/6	5.7	141	5.965	300
4	Hauchab, N Lebanon $\beta_p 3$ (157—162)	347.6	+ 46.1	5/6	3.9	387	4.990	150
5	Saidnaya, N Lebanon $\beta_p 3$ (163—169)	6.4	+ 48.5	7/7	7.1	73	6.917	300
6	Tlail, N Lebanon $\beta_p 3$ (170—174)	81.7	+ 58.4	3/5	3.4	1322	2.998	300
7	Amriye, N Lebanon $\beta_p 3$ (175—180)	182.2	− 35.7	5/6	5.0	233	4.983	150
8	Baghdadi, N Lebanon $\beta_p 3$ (185—188)	172.2	− 29.1	4/4	6.8	186	3.984	300

TABLE II (continued)

Sites		D	I	Sampling*	α_{95}	K	R	Demagn. (Oe)
9	Bire, N Lebanon $\beta_p 3$ (189—195) flexure	43.0	+ 51.0	7/7	8.5	51	6.883	150
10	Ouadi Chadra, N Lebanon $\beta_p 3$ (196—200)	192.1	− 49.5	5/5	7.1	116	4.965	300
11	Same as 10, 20 m lower $\beta_p 3$ (201—206)	21.8	+ 49.5	6/6	9.2	54	5.908	150
12	Kousba, N Lebanon (207—211) $\beta_p 1$ intercalated in Pleistocene limestone	193.2	− 45.9	5/5	5.3	213	4.981	150
13	Merj ez-Zhour, S Lebanon (101—105; 255—257) $\beta_p 3$?	357.9	+ 53.2	7/8	4.6	177	6.966	150–300
13a	Same as 13, collected in 1963	352.0	+ 47.2	9/9	7.7	46	8.827	85
14	Tell Ras Hirbaya, S Lebanon (258—263) $\beta_p 3$?	345.3	+ 43.2	4/6	17.3	20	4.804	300
15	Hasbaya, S Lebanon (264—269) $\beta_p 3$?	348.9	+ 22.2	5/6	19.0	17	4.776	300
16	El Mhaidthe, S Lebanon (316—321) $\beta_p 3$?	349.6	+ 40.0	4/6	9.0	105	3.971	150
Overall mean direction (sites 6, 13a excluded)		1.6	+ 46.2	—	7.7	28	13.533	—
Virtual pole position		169.3 E 10.7 W	88.2 N 88.2 S		dp = 6.3 dm = 9.8			

* Number of samples used in computation/number of samples collected.

** The rift boundary fault is believed to have a left lateral displacement which might account for this deviant direction.

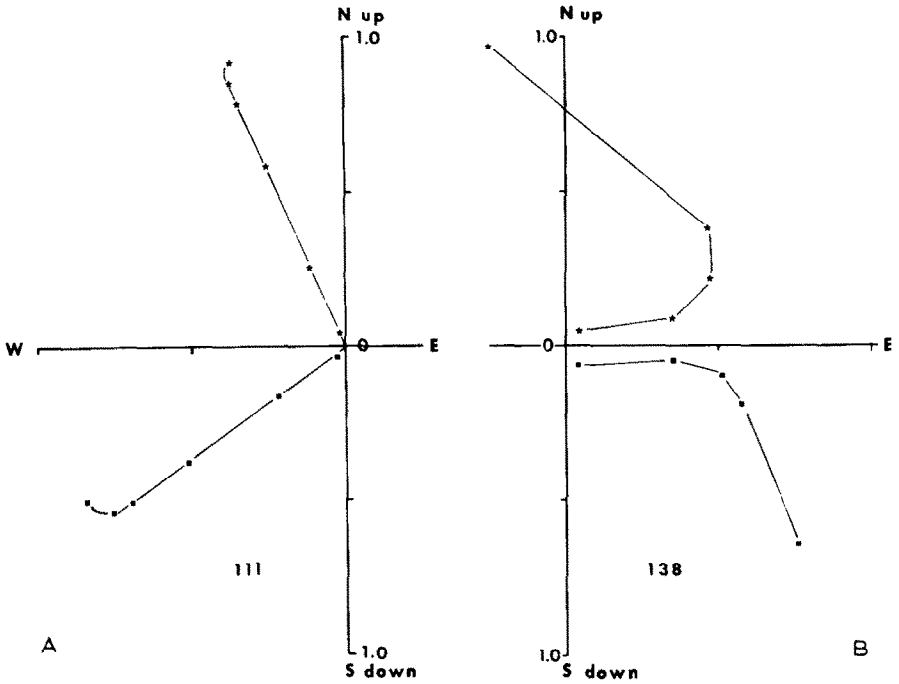


Fig. 4. Plots of magnetic vector during alternating field demagnetization. Numerical scale indicates normalized intensity (residual intensity/NRM intensity) measured radially from the origin. Stars indicate changes in declination; squares, changes in inclination. A. A Pliocene sample. B. A Jurassic sample.

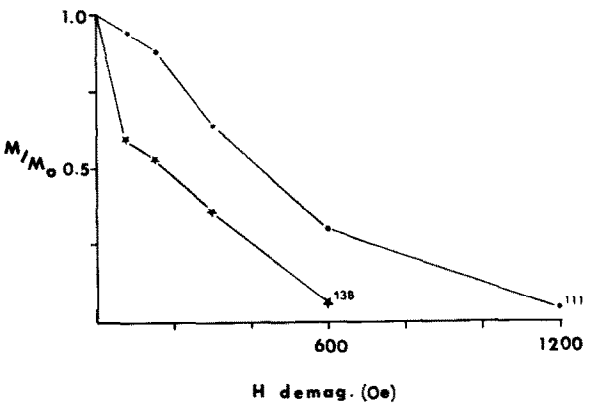


Fig. 5. Plots of normalized intensity against intensity of demagnetizing field. Same samples as in Fig. 4.

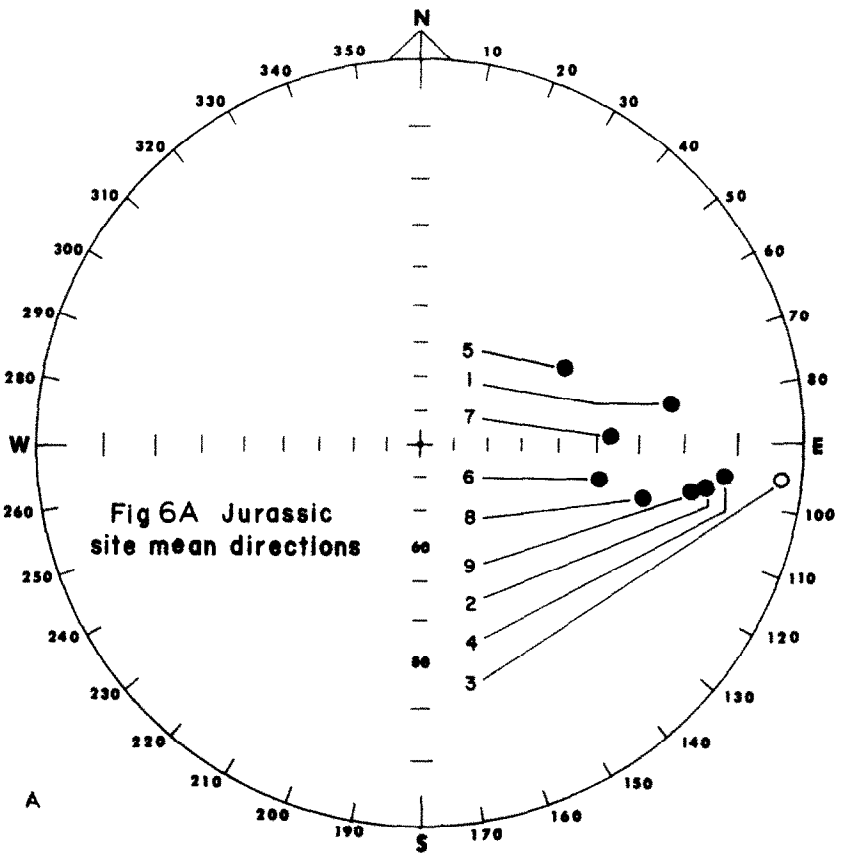


Fig. 6A. Jurassic site mean directions of magnetization after magnetic cleaning.

standpoint these anomalies do not affect the mean direction of magnetization.

DISCUSSION OF RESULTS

Both normal and reversed directions were found in the Cretaceous and Pliocene samples, but only a single direction in the Jurassic group. Assuming that their Aptian samples have normal polarity, we follow Van Dongen et al. (1967) in considering this Jurassic direction, and all the Southeast Neocomian directions, as reversed, so minimizing the virtual pole shift during the Kimeridgian—Aptian interval. In the tabulation by McElhinny and Bürek (1971) the field is reversed during part of Kimeridgian time, and Helsley and Steiner (1969) report two reversals in the Neocomian.

The virtual pole positions are listed in Table III together with others from the Middle East and comparable poles from India and Africa. The low latitudes of the Jurassic and Cretaceous virtual poles from Lebanon are regarded

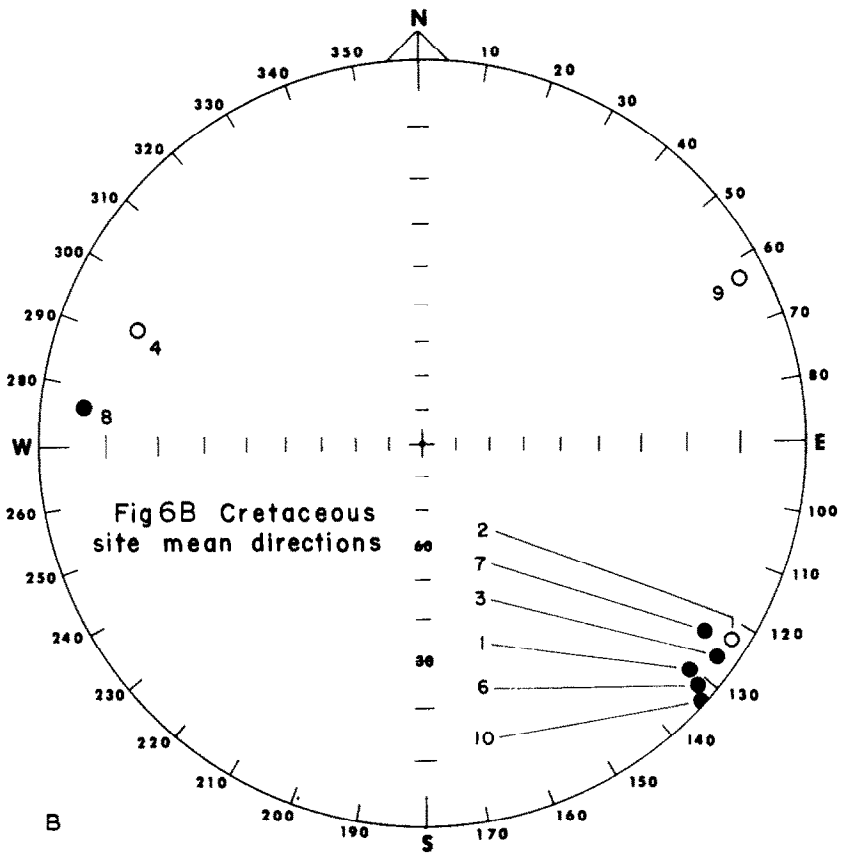


Fig. 6B. Cretaceous site mean directions of magnetization after magnetic cleaning.

as real, since the time spread of the results reported is too great for them to represent an excursion of the geomagnetic field such as reported by Watkins and Cambray (1971) among others. Taken at their face value they indicate a progressive anticlockwise rotation of Lebanon during late Mesozoic and perhaps early Cenozoic times, accompanied by a northward change in latitude (Fig. 7). These movements were apparently concluded before the extrusion of the Pliocene basalts. The same general movement is also indicated for Cyprus by the results of Vine and Moores (1969), and it thus seems possible that the two masses behaved as a single tectonic unit independent of Turkey to the north (Van der Voo, 1968), a conclusion consistent with McKenzie's (1970) postulate of an independent Turkish plate.

While the Lebanese virtual pole positions are closer to those of India than to the African Jurassic—Cretaceous pole (see Table III), attempts to attach Lebanon to the Indian plate meet with such overwhelming difficulties that the agreement must be regarded as fortuitous. On the contrary, the continu-

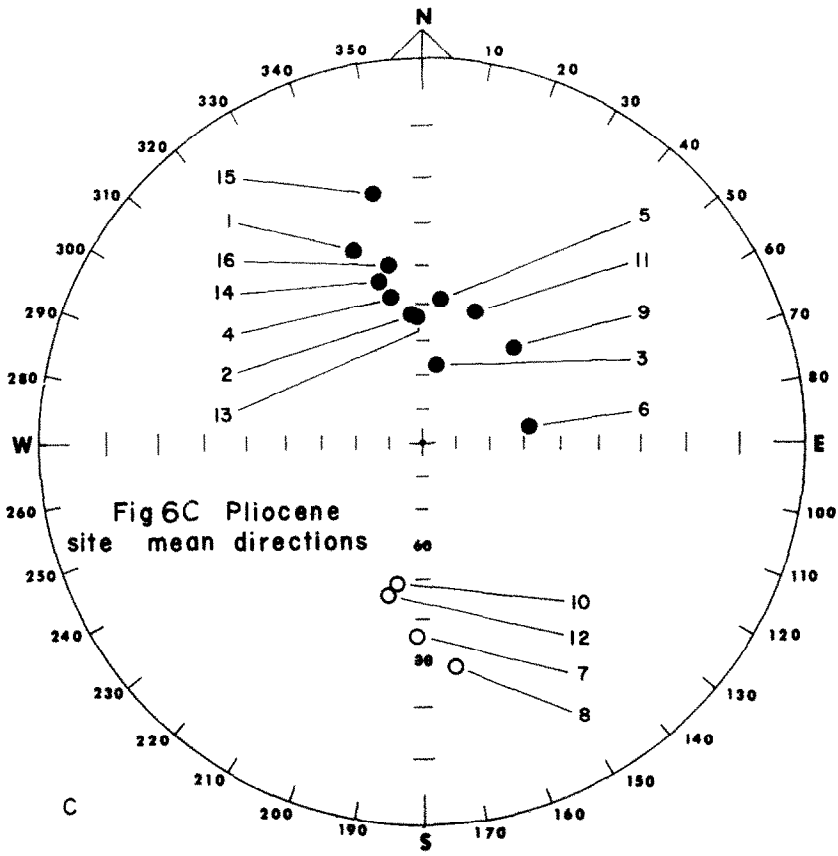


Fig. 6C. Pliocene site mean directions of magnetization after magnetic cleaning.

ity of the Nubian Facies from Africa to Lebanon indicates that Lebanon has for long been a part of the African plate; so the Lebanese and African poles have to be reconciled. Since movement of Arabia relative to Africa has been limited to the opening of the Red Sea, the reconciliation requires a sinistral rotation of Lebanon relative to the Arabian plate of almost 70° since the eruption of the Kimeridgian basalts (see Fig. 7). The position of the Neocomian virtual pole for Lebanon (Fig. 7) implies a post-Neocomian anticlockwise rotation of about 40° relative to Africa: say 35° relative to Arabia, allowing 5° between Arabia and Africa to open the Red Sea. It would be satisfying to find field evidence of some such movement as the paleomagnetic results indicate; but the present political situation in the Levant makes it difficult to pursue regional studies across international boundaries. Working in central Lebanon Kana'an (1966) concludes that the Grès du Liban is a fluviodeltaic deposit, and his observations on the cross-bedding (plotted in Fig. 8) suggest a generally westward paleoslope with a strand line

TABLE III

Virtual pole positions from Jurassic and Cretaceous rocks of the Lebanon and adjacent areas

Location	Formation	Virtual pole	dp	dm	α_{95}	Reference
Cretaceous						
Lebanon	Neocomian basalts	75 W 25 N	4.5	9.0	9.0	this paper
Lebanon	Aptian basalts and limestones	78 W 38 N	—	—	5.5	Van Dongen et al. (1967)
Turkey	Gümüshane sandstones and volcanics	81.5 W 60.5 N	—	—	5.5	Van der Voo (1968)
Turkey	Niksar volcanics	64 W 51 N	—	—	16	Van der Voo (1968)
Cyprus	gabbro and Precampanian pillow lavas	46.8 W 18.7 N*	—	—	—	Vine and Moores (1969)
S Israel	Lower Cretaceous lava	95 W 53 N	—	—	—	Helsley and Nur (1970)
N Israel	Senonian lava	96 W 42 N	—	—	—	Helsley and Nur (1970)
India	Rajmahal traps (Albian, 100 m.y.)	62 W 3 N	—	—	6.0	McDougall and McElhinny (1970)
India	Sylhet traps (? Lower Cretaceous)	60 W 16 N	—	—	8.0	Athavale et al. (1963)
India	Tirupati Sandstones (M—U Cret.**)	73 W 30 N	—	—	5.0	Verma and Pullaiah (1967)
Africa	Lupata Volcanic Series (106—111 m.y.)	100 W 62 N	3	5	4.0	McElhinny et al. (1968)
Africa	Mlanje Massif syenite (116—128 m.y.)	98 W 60 N	9	13	12.0	McElhinny et al. (1968)
Jurassic						
Lebanon	Kimeridgian basalts	66 W 2 S	5.9	11.2	10.6	this paper
Lebanon	Kimeridgian basalts and tuffs	60 W 1 S	—	—	3	Van Dongen et al. (1967)
India	Parsora Sandstone (Upper Triassic)	55 W 30 N	—	—	6.0	McDougall and McElhinny (1970)
India	Pachmarhi Beds (Upper Triassic)	50 W 10 N	—	—	10.0	McDougall and McElhinny (1970)
Africa	Mateke Hills Ring Complex (177 ± 4 m.y.)	100.5 W 58.5 N	—	—	8.5	McElhinny et al. (1968)

* Computed from $D = 280$, $I = +34$ using estimated site coordinates 33E 35N.
** Discussion of age in McDougall and McElhinny (1970).

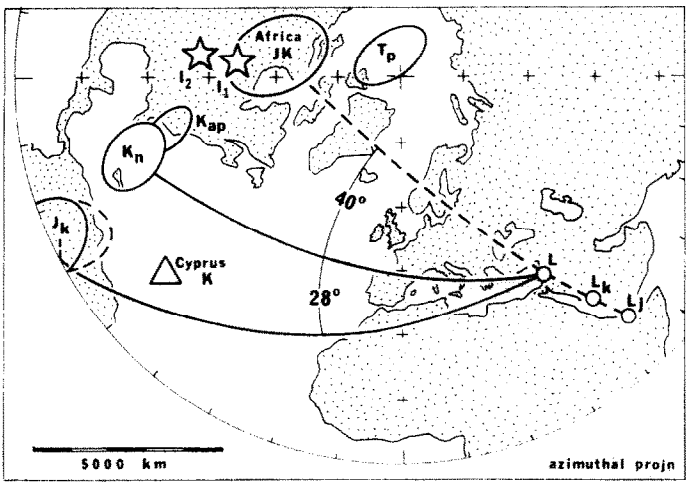


Fig. 7. Virtual pole positions from Lebanese rocks compared with the Jurassic—Cretaceous pole for Africa (see Table III). J_k = Kimeridgian (this paper); K_n = Neocomian (this paper); K_{ap} = Aptian (Van Dongen et al., 1967); T_p = Pliocene (this paper). Lower (I_1) and Upper (I_2) Cretaceous poles for Israel and pre-Campanian pole for Cyprus are also shown. L_j, L_k = Upper Jurassic and Lower Cretaceous projections of Lebanon on a paleomeridian drawn from the African pole; L=present position of Lebanon. Discussion in text.

running roughly north and south; and a petrographic study by Wakim (1968) supports the widely-held belief that the source area of the Grès du Liban lay on the Arabian shield to the south. Kana'an's and Wakim's observations fit the general paleogeography of the region just as well after the reconstruction proposed here (Fig. 8b) as they do with Lebanon in its present-day position.

The rotation indicated by our results (Fig. 7) has occurred during the time (Cretaceous) when Africa was moving away from North America while Europe and North America were still attached to one another, and Africa consequently moved eastward relative to Europe (See Phillips and Forsyth, 1972; Hsü, 1971). This is the state of affairs invoked by Argand (1924) to account for the "puzzling geography of the western Mediterranean", and refined and placed in its correct chronology by Hsü (1971). The anticlockwise rotations of Spain, Sardo-Corsica and Italy inferred by Argand have been verified paleomagnetically in recent years by Van der Voo (1967), Nairn and Westphal (1968), Zijderveld et al. (1970), and Zijderveld and De Jong (1969). The present results from Lebanon agree well with those from an earlier sampling by Van Dongen et al. (1967) and amplify them by adding a Neocomian virtual pole between the Kimeridgian and Aptian ones. We therefore conclude that in the eastward march of Africa relative to the Eurasian plate, Lebanon underwent a rotation similar to those of the west European microcontinents. The low paleolatitudes are consistent with Hsü's postulate of a northward movement of Africa during the past 40 million years, but it appears (see Fig. 7) that there has also been some northerly displacement relative to Africa itself.

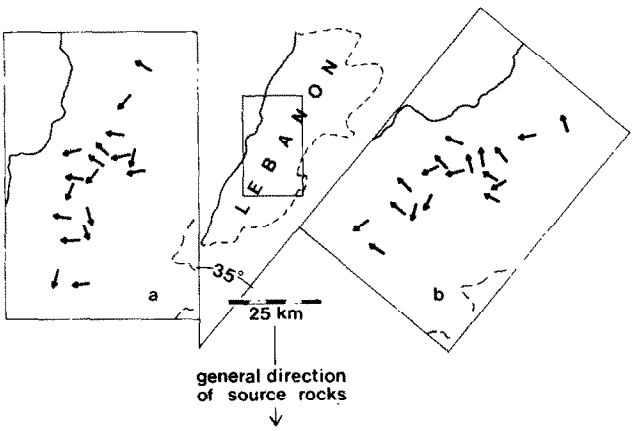


Fig. 8. Orientation of Lebanon relative to supposed source area of the Grès du Liban (Neocomian). *a* = present; *b* = Neocomian, as indicated by the paleomagnetic results. Arrows represent vector means of cross-bedding dip directions at 18 sites in the Grès du Liban (data from Kana'an, 1966).

If an extensive rotation of Lebanon has occurred, where lies the tectonic boundary between Lebanon and the rest of the African plate? Helsley and Nur (1970) have sampled three sites (22 samples) in a Lower Cretaceous lava in the Negev (southern Israel) and one (15 samples) in a Senonian lava in the Carmel area (northern Israel); the resulting virtual poles are shown here on Fig. 7. The pole from southern Israel lies on the edge of the confidence oval for the African pole, implying that this area has not participated in the rotation of Lebanon; the pole from Carmel is close by, a little to the south. This may mean either that the Carmel area was also separate from the Lebanon, or that it was attached to Lebanon but the rotation was almost over by Senonian time: that it did not last very long after the Mesozoic is suggested by the continuity into Lebanon of the Dead Sea Rift, which was already sketched out in the Oligocene. A more extensive sampling in Israel might help considerably in the search for a southern boundary to the Lebanese sub-plate.

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